## Design of Two Stage Miller Compensated Operational Transconductance Amplifier(OTA) for the given specifications.

### OTA Specifications

- Open loop gain$\geq$ 60 $dB$
- Slew rate = $20V/\mu s$
- Gain Bandwidth Product = 10 MHz
- Phase Margin = 60°
- Input common mode range=0.9 V to 1.6 V
- Load capacitor, $C_L$= $10pF$

### Design Procedure

Following transistors parameter are known for the design: $V_{tn}$=$|V_{tp}|$ = 0.4 V, $K'_n$= 218 $\mu AV^{-2}$, $K'_p$= 90 $\mu AV^{-2}$.

Fig. 1 shows the circuit level diagram of 2-stage Miller Compensated OTA. The block level representation of

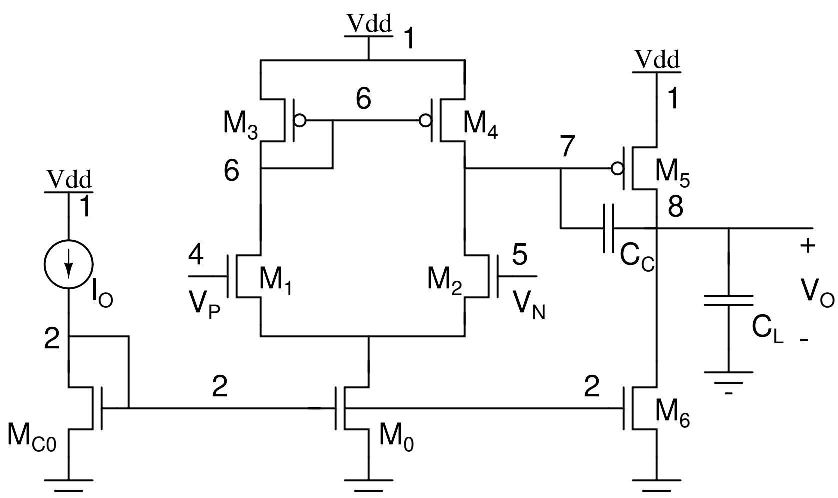


Figure 1: Schematic of 2-stage Miller compensated OTA .

the two stage Miller compensated OTA is represented by Fig. 2. The transfer function H(s) is given by

$$\frac{V_o(s)}{V_d(s)} = \frac{K\,(1 - s/z_1)}{(1 + s/p1)(1 + s/p2)} \tag{1}$$

where,
$K = (g_{m1}g_{m2}/g_{o1}g_{o2})$ is the small signal DC gain,
$p_1 = (g_{o1}g_{o2}/C_c g_{m2})$ is the dominant pole,
$p_2 = (g_{m2}/C_L)$ is the second pole,
$z_1 = (g_{m2}/C_c)$is the right half plane zero.

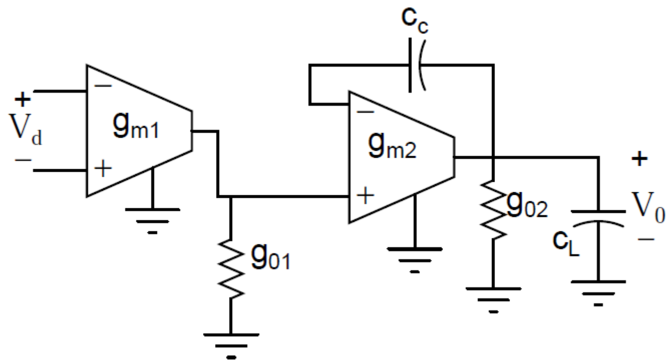


Figure 2: Block level representation of 2-stage Miller compensated OTA .

**Evaluation of $C_c$ from $C_L$**

To obtain a phase margin of 60°, it is required that the compensating capacitor $C_c$ satisfy this following condition, i.e

$$C_c = 0.2C_L \tag{2}$$

Hence, $C_c$ is evaluated to 2 pF.

**Evaluation of $I_o$ from slew rate**

The slew rate of 2-stage Miller compensated OTA is given by,

$$SR = I_o/C_c \tag{3}$$

Hence $I_o$ is evaluated to 40 $\mu A$.

**Evaluation of $G_{m1}$**

We have,

$$GBW = \frac{G_{m1}}{C_c} rad/s \tag{4}$$

Hence, $G_{m1}$ is evaluated to 125.7$\mu S$.

**Evaluation of $(W/L)_1$**

We have,

$$(W/L)_{1,2} = \frac{{g_{m1}}^2}{I_o K_n{}'} \tag{5}$$

The $(W/L)_{1,2}$ is evaluated to 1.811. With $L = 1\mu m$, $W_1$ ≈2$\mu m$.

**Evaluation of $(W/L)_0$ from $V_{BIAS(MIN)}$=0.9 V**

It is required that,

$$V_{BIAS(Min)} \geq V_{GS1} + V_{dsat0}$$

$$V_{BIAS(Min)} \geq V_{dsat1} + V_{tn} + V_{dsat0}$$

$$V_{dsat0} = V_{BIAS(Min)} - V_{tn} + V_{dsat1}$$

$$V_{dsat0} = V_{BIAS(Min)} - Vtn - \sqrt{\frac{I_0}{K_n'(W/L)_1}}$$

$V_{dsat0}$ is evaluated to 0.2 $V$. Hence $(W/L)_0$ is evaluated from $V_{dsat0}$.

$$(W/L)_0 = \frac{I_o}{V_{dsat0}^2 K_n{}'}$$

The $(W/L)_0$ is evaluated to 9. With $L = 1\mu m$, $W_0 \approx\ 9\mu m$.

**Evaluation of $(W/L)_{3,4}$ from $V_{BIAS(MAX)}$=1.6 V**

It is required that $M_{1,2}$ must be in saturation for $V_{BIAS(MAX)}$.

$$V_{D1} \geq V_{BIAS(MAX)} - V_{tn}$$

$$V_{dsat3} = V_{DD} - V_D - |V_{tp}|$$

$V_{dsat3}$ is evaluated to 0.2 $V$. Hence $(W/L)_3$ is evaluated from $V_{dsat3}$.

$$(W/L)_3 = \frac{I_o}{V_{dsat3}^2 K_p{}'}$$

The $(W/L)_3$ is evaluated to 11.1. With $L = 1\mu m$, $W_0 \approx\ 11\ \mu m$.

**Evaluation of $(W/L)_5$ and $(W/L)_6$**

It is required that the $2^{nd}$ stage must carry current of approximately 5 times the current in stage1 such that the second pole is far from the dominant pole. So the required current in $2^{nd}$ stage is 200 $\mu A$. From this condition, $W_6 \approx 5 * W_0$ and $W_5 \approx 10 * W_3$. The dimensions of $M_5$ and $M_6$ are evaluated to 110 $\mu m$ and 45 $\mu m$ respectively.

The DC gain calculated from the above evaluated values is around 2000=60 dB. Thus meet the given specification

The dimensions of all the transistors with L=1 $\mu m$ is tabulated in Table. 1.

| Transistors | Width ($\mu m$) |
|---|---|
| $M_0$,$M_{C0}$ | 9$\mu$m |
| $M_1$ | 2$\mu$m |
| $M_2$ | 2$\mu m$ |
| $M_3$ | 11$\mu m$ |
| $M_4$ | 11$\mu m$ |
| $M_5$ | 110$\mu m$ |
| $M_6$ | 45$\mu m$ |

Table 1: Dimensions of all the transistors from the design.

## Zero Cancellation with $R_c$

From the transfer function analysed before, there is a zero present in right half plane, which degrades the phase margin. The phase margin is improved by cancelling this zero by placing a resistor $R_c$ in series with $C_c$ (as seen in Fig. 3), where $R_c\ > 1/Gm2$. Choose $R_c = 1\ K\Omega$.

### 0.0.1 Simulation setup to determine the loop gain and phase margin

A large test resistor $R_{test}$ and test capacitor $C_{test}$ is connected as shown in the Fig. 5. The setup maintains required DC operating points with the feedback loop open for AC analysis.

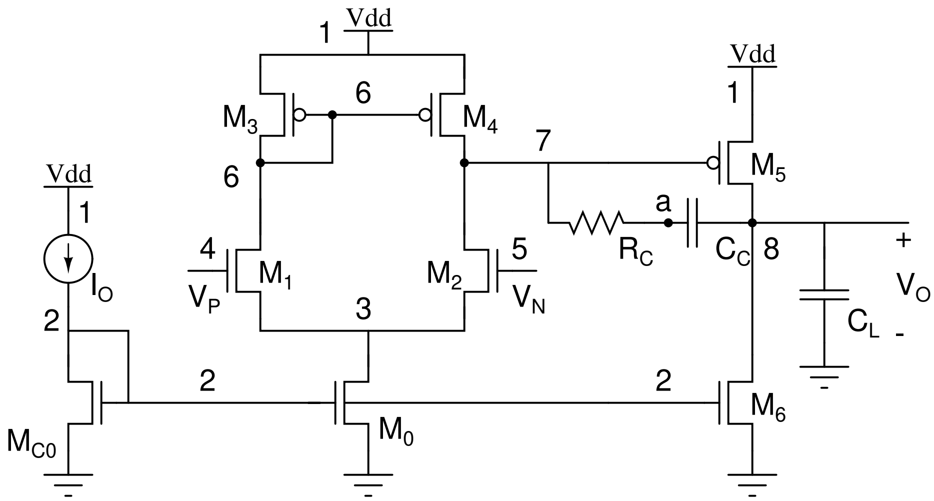


Figure 3: Schematic of 2-stage Miller compensated OTA with zero cancellation.

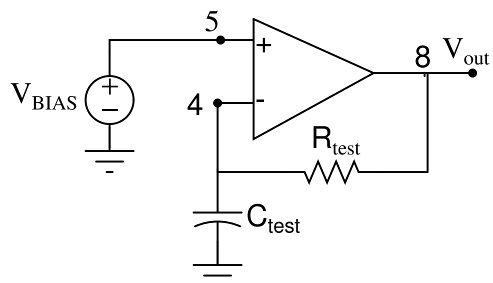


Figure 4: Simulation setup to determine loop gain and phase margin

# Exercise

1. Observe the phase response for the above OTA for different values of $C_c$ and $R_c$.
2. Apply a small signal step input of 5 mV over the bias voltage of 1.25 $V$ with the unity gain configuration (as shown in Fig. **??**). Observe the step response. Repeat it for different values of $C_c$ and $R_c$.

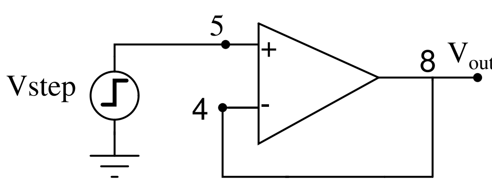


Figure 5: OPAMP in unity gain configuration

3. Apply large signal step input ranging from 0.9 V to 1.6 V. Determine the input slew rate.